
Solver-Pinned Distillation for Strategic Reasoning in Small LLMs

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Abstract

1 Frontier LLMs reason fluently about non-cooperative games but play poorly against
2 strategic opponents: against tit-for-tat in iterated prisoner’s dilemma, Haiku 4.5
3 averages 1.50 cumulative reward per trajectory, the same as the Qwen 2.5–3B base
4 model. We investigate whether DPO from solver-supervised labels can close this
5 gap. Our recipe, $A+\beta$, replaces frontier-model action labels with the provably-
6 optimal best response from analytical game-theoretic solvers (closed-form for
7 matrix games, BNE for first-price auctions, take-the-rest for multi-issue bargaining),
8 uses a frontier model only to paraphrase reasoning under a no-leakage posterior-EV
9 prompt, and filters pairs by counterfactual horizon rollout against deterministic
10 opponents. On a 12-environment evaluation suite (8 in-domain 2-player games + 4
11 held-out structurally distinct games), Qwen 2.5–3B trained with $A+\beta$ closes 93%
12 of the held-out cumulative-reward gap to Haiku 4.5 (+1.91 vs frontier +2.05, vs
13 base −1.45) and lifts tit-for-tat-opponent reward from 1.50 to 14.06 (+12.5). We
14 further show that two specialist adapters, one trained for diversity coverage and
15 one for horizon-aware strategic play, can be merged by simple parameter averaging
16 to recover combined capabilities without any additional fine-tuning, matching the
17 reweighted retrain to within 0.16 reward.

18 1 Introduction

19 Recent benchmarks have established that large language models (LLMs) underperform game-theoretic
20 solution concepts in non-cooperative games beh [2025], gtb [2024], even when given chain-of-thought
21 room to reason. The natural next question is what training-time intervention closes that gap, and
22 the natural training-time intervention is direct preference optimization (DPO) Rafailov et al. [2023]
23 on game-play trajectories: collect rollouts of an agent playing some game, label the trajectories by
24 realised cumulative reward, and pair high-reward traces against low-reward ones.

25 This paper reports an investigation of that intervention. The intervention works, but not for the reason
26 the standard DPO recipe predicts. Three findings drive the paper.

27 **(1) Frontier LLMs play strategic opponents poorly, not just small ones.** On iterated prisoner’s
28 dilemma against tit-for-tat, Haiku 4.5 averages 1.50 cumulative reward per trajectory — identical to
29 the Qwen 2.5–3B base model. Llama 3.3–70B scores 1.50 as well. Frontier coupling rates between
30 stated expected-value computations and chosen actions cluster between 0.69 and 0.85; on held-out
31 games the range drops to 0.30–0.79. Scale alone does not close the gap. We use this as the empirical
32 entry point: if Haiku and Qwen-3B are comparably weak against TFT, frontier-as-teacher distillation
33 cannot be the right recipe.

34 **(2) Solver-pinned best responses outperform frontier-played actions as DPO labels.** We intro-
35 duce $A+\beta$ (Algorithm-pinned with β -counterfactual filter), a data-construction recipe that replaces

the frontier’s recorded action with the provably-optimal best response from an analytical solver — closed-form for 2-player matrix games, the symmetric BNE bid for first-price auctions, the take-the-rest proposal for multi-issue bargaining — and uses the frontier only as a *paraphraser* of reasoning trace conditioned on that solver-pinned action. A counterfactual horizon-rollout filter (Option β) prunes solver actions whose myopic optimality would fail under the realised opponent’s reaction. Trained on 1338 pairs mixing matrix-game, auction, negotiation, and horizon-aware data, Qwen 2.5–3B closes 93% of Haiku 4.5’s held-out CR margin (+1.91 vs +2.05, vs base −1.45) and lifts TFT-opponent CR from 1.50 to 14.06 (Table 1).

(3) Two specialist adapters trained on disjoint data slices merge by parameter averaging without retraining, matching the reweighted-retrain alternative within sampling noise. We observe a specific forgetting trade-off mid-investigation: one mix (AUX, 613 pairs, auction+negotiation-heavy) wins on held-out reward but loses ID strategic play, while another (ALL, 1338 pairs, horizon-aware-PD heavy) wins ID strategic play but loses auction transfer. Two interventions resolve this: (a) reweighting the training mix to upsample the underrepresented subdistribution 4 $\times$, and (b) averaging the AUX and ALL LoRA adapters $W_{\text{merge}} = 0.5 W_{\text{AUX}} + 0.5 W_{\text{ALL}}$ with no retraining. The two recover the same combined capability profile to within 0.16 cumulative reward, suggesting the two specialists learned linearly composable LoRA subspaces rather than conflicting representations.

Contributions. We make four contributions.

1. **A+ β method (§2).** A DPO data-construction recipe that decouples action labels (analytical solver) from reasoning labels (frontier paraphrase under a no-leakage posterior-EV prompt), and prunes pairs by counterfactual horizon rollout against deterministic opponents. We also present a horizon-aware variant that uses dynamic-programming rollout to assign credit beyond stage-game best response.
2. **12-environment evaluation suite (§3).** Eight in-domain 2-player games (iterated prisoner’s dilemmas with three payoff variants, Stag Hunt, multi-issue bargaining, Battle of Sexes, Matching Pennies, tic-tac-toe) and four structurally distinct held-out games (first-price auction, divide-the-dollar, K -level p -beauty, one-shot prisoner’s dilemma stage). Frontier coverage measured at full eval scale ($n=12$ episodes per env, 4 frontier models + Qwen 2.5–3B base). The benchmark and scores are independent of the rest of our contributions.
3. **Empirical findings (§3).** A+ β -trained Qwen 2.5–3B closes 93% of the held-out CR gap to Haiku 4.5; lifts strategic-opponent CR from 1.50 to 14.06; matches or beats Haiku on three of eight ID games (pd-high-temptation, stag-hunt, ipd-stage). No 3B variant matches Haiku’s aggregate ID coupling 0.85 ceiling.
4. **Specialist-adapter LoRA averaging (§2, §3.5).** Two A+ β adapters trained on disjoint data slices (AUX, ALL) recover combined capabilities under simple parameter averaging, matching the reweighted retrain alternative to within sampling noise. To our knowledge, this is the first demonstration of LoRA-merge equivalence for strategic-game DPO specialists.

What this paper does not claim. A+ β -trained 3B does not beat Haiku 4.5 on aggregate ID cumulative reward (13.93 vs 14.60) or on aggregate ID coupling (0.68 vs 0.85). On games where Haiku already exploits weaker opponents (e.g. pd-classic against always-cooperate) the trained model loses 1–2 reward points. A+ β closes the held-out and strategic-opponent gaps; the in-domain coupling gap remains.

2 Method

We describe A+ β , a data-construction recipe for direct preference optimization on game-play trajectories. The recipe replaces the frontier model’s action label with an analytical solver’s best response, uses the frontier only to paraphrase reasoning under a no-leakage prompt, and filters pairs by counterfactual horizon rollout against deterministic opponents. §2.1 fixes notation, §2.2 describes the solver-pinned best-response operator across our three game families, §2.3 describes the no-leakage frontier paraphrase, §2.4 describes the counterfactual filter and its horizon-aware variant, and §2.5 describes specialist-adapter LoRA averaging. Figure 1 sketches the data-construction pipeline.

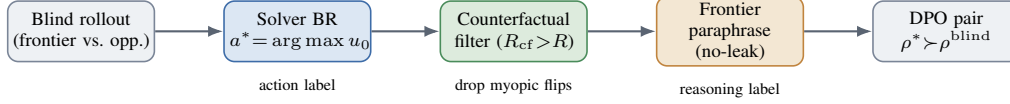


Figure 1: $A+\beta$ data construction. A blind rollout produces the rejected side. The analytical solver supplies the *action* label (provably-optimal best response to the realised opponent move); a counterfactual horizon rollout drops flips whose myopic optimality would not improve episode reward; the frontier model supplies the *reasoning* label under a no-leakage posterior-EV prompt. The chosen side pairs the solver action with the frontier reasoning.

2.1 Notation and setup

We work with two-player games $G = (\mathcal{A}_0, \mathcal{A}_1, u_0, u_1)$ under either of two modes. *One-shot* games (auctions, multi-issue bargaining, single-stage matrix games) terminate after one joint action. *Repeated* games of horizon T (iterated prisoner’s dilemma variants, repeated stag-hunt) play the same stage game T times, with each player observing the joint history before each round. We focus on player 0; player 1 is the opponent.

A trajectory $\tau = (x, \rho, \mathbf{a})$ contains a game context x , a chain-of-thought reasoning string ρ_t at each round t , and an action sequence $\mathbf{a} = (a_t^0)_{t \leq T}$. The realised cumulative reward is $R(\tau) = \sum_t u_0(a_t^0, a_t^1)$ where a_t^1 is the opponent’s action at round t , sampled from a deterministic heuristic policy π_{opp} during training-data generation.

DPO Rafailov et al. [2023] requires preference pairs (x, y_w, y_l) where y_w is preferred to y_l given context x . For game-play trajectories, the natural preference signal is realised cumulative reward, but on a single rollout this signal couples three sources of variance: which round the model reasoned well, which round it reasoned poorly, and how the opponent reacted. $A+\beta$ addresses each separately.

2.2 Solver-pinned best response

For each round t in a generated trajectory, we substitute the frontier model’s recorded action with the provably-optimal best response computed by an analytical solver, conditioned on the opponent’s actually-realised action a_t^1 :

$$\text{BR}_0(a_t^1) = \arg \max_{a \in \mathcal{A}_0} u_0(a, a_t^1). \quad (1)$$

We implement Eq. 1 for three game families. For **2-player matrix games**, the solver enumerates $|\mathcal{A}_0|$ own-action options and returns the argmax over the recorded opponent column; computation is $O(|\mathcal{A}_0|)$. For **first-price sealed-bid auctions** with private values $v_0, v_1 \sim \text{Unif}[0, v_{\max}]$, the realised best response given the opponent’s bid b_1 is the smallest legal bid strictly above b_1 if $v_0 > b_1 + \epsilon$, else 0 (lose deliberately). For **multi-issue bargaining** with capacity c across three issues and weights w_0 , the realised best response given the opponent’s proposal $\mathbf{p}_1$ is the take-the-rest proposal $p_0[i] = c - p_1[i]$, which maximises u_0 subject to feasibility $(p_0[i] + p_1[i] \leq c \text{ for all } i)$.

Why solver-pinned, not frontier-pinned. An empirical observation motivates this choice. Frontier LLMs produce sophisticated reasoning traces (posterior beliefs, expected-value computations) but their actions deviate from the argmax of their own stated EVs at rates of 15–45% depending on game and round (Appendix 5). Pinning DPO labels to the frontier’s action therefore locks in the decoupling. Pinning to the solver gives a provably correct action label with computational cost bounded by $|\mathcal{A}_0|$ for matrix games and the legal-bid grid for auctions.

2.3 No-leakage frontier paraphrase

Given the round’s solver-pinned action a^* , we use a frontier model (Bedrock Haiku 4.5) only to write the chain-of-thought reasoning ρ_t that would justify a^* . The reasoning prompt faces a subtle constraint: at deployment, the small model will see *history up to round t* but not the current round’s opponent action. Reasoning that says “the opponent played D this round, therefore I defect” would leak information that does not exist at inference time.

123 We solve this with a posterior-EV reframing of the paraphrase prompt. The frontier is told the
 124 opponent’s action and the solver’s BR out-of-band, but is instructed to write reasoning expressible
 125 from history alone:

126 SYSTEM. You are documenting strategic decisions . . . The other player’s current-round
 127 move is NOT observable at decision time. We will tell you (out-of-band, for your reference)
 128 the opponent’s actual move that round and the solver-computed best response, but your
 129 written reasoning must NEVER reference the current-round opponent move as observed.
 130 Treat it as belief-state reasoning under uncertainty. . . . Hard rules: (1) NEVER write
 131 phrases like “opponent played C this round”; (2) The [EV] arithmetic must average over
 132 your posterior types — not condition on the unrevealed action; (3) Your [Decision] must
 133 match the solver-pinned best response we provide; build the posterior to justify that action.

134 Audited across 688 oracle rounds in our final batches, zero round-1 traces leak the current opponent
 135 action and 1% of all rounds match a heuristic leakage regex (manual inspection finds all 1% are false
 136 positives discussing prior rounds; see Appendix 5).

137 2.4 Counterfactual horizon filter

138 The solver-pinned action of §2.2 is myopic: “defect against any cooperator now, regardless of future
 139 rounds.” In repeated games against reciprocity opponents (tit-for-tat, grim trigger), myopic defection
 140 on round 1 triggers permanent retaliation and tanks long-run reward. Pairing every solver flip as a
 141 chosen example would teach defection-bias.

142 We filter solver-flipped pairs by counterfactual rollout. Given a recorded blind trajectory τ_{blind} and a
 143 candidate flip at round t with action a^* , we compute the counterfactual cumulative reward:

$$R_{\text{cf}}(t, a^*) = \sum_{s < t} u_0(a_s^0, a_s^1) + u_0(a^*, a_t^1) + \sum_{s > t} u_0(a_s^0, \pi_{\text{opp}}(\mathbf{h}_s^*)) \quad (2)$$

144 where $\mathbf{h}_s^*$ is the counterfactual history at round s including the flip, and π_{opp} is the deterministic
 145 opponent’s recorded policy. Because our training opponents are deterministic heuristics (π_{opp} is
 146 reconstructible from its name), this counterfactual is exact, not estimated. We emit a DPO pair
 147 ($\rho_t^{\text{oracle}} \succ \rho_t^{\text{blind}}$) only when $R_{\text{cf}}(t, a^*) > R(\tau_{\text{blind}})$.

148 **Horizon-aware variant.** Eq. 2 holds the small model’s post-flip play fixed at the blind-recorded
 149 actions. A more ambitious counterfactual asks: *what if, after the flip, the small model played optimally*
 150 *for the remainder of the episode?*

151 We compute this by full dynamic-programming rollout. Define $V(s, \mathbf{h}_s)$ as the maximum attainable
 152 cumulative reward from state s given history $\mathbf{h}_s$:

$$V(s, \mathbf{h}_s) = \max_{a \in \mathcal{A}_0} \left[u_0(a, \pi_{\text{opp}}(\mathbf{h}_s)) + V(s+1, \mathbf{h}_s \oplus (a, \pi_{\text{opp}}(\mathbf{h}_s))) \right] \quad (3)$$

153 with $V(T, \cdot) = 0$. The horizon-aware counterfactual reward is $R_{\text{cf}}^{\text{HA}}(t, a^*) = \sum_{s < t} u_0(a_s^0, a_s^1) +$
 154 $u_0(a^*, a_t^1) + V(t+1, \mathbf{h}_{t+1}^*)$. For matrix games with $|\mathcal{A}_0| = 2$ and $T \leq 10$, full enumeration of 2^{T-t}
 155 branches is tractable; we did not need to prune. The horizon-aware variant flips DP-correctness over
 156 a deterministic opponent into a training-time signal.

157 **Connection to GRPO.** GRPO Shao et al. [2024] addresses the same credit-assignment problem
 158 in repeated games by sampling K on-policy rollouts and group-normalising terminal rewards. Our
 159 horizon-aware counterfactual replaces this K -sample estimate with a single deterministic DP pass
 160 when the opponent’s policy is known. The tradeoff is exactness vs. generality: GRPO works against
 161 any opponent, our DP works only when the opponent is reconstructible from its name. For the
 162 deterministic-heuristic training regime, we recover what GRPO would converge to in the limit, with
 163 no rollout cost.

2.5 Specialist-adapter LoRA averaging

A practical observation from §3.5 motivates a final component. $A+\beta$ training on different data mixes produces adapters with complementary strengths: AUX (auction+negotiation-heavy) generalises out-of-domain; ALL (horizon-aware-PD-heavy) wins on strategic in-domain play. The two adapters are trained with the same LoRA configuration ($r = 16$, $\alpha = 32$, target modules identical) on the same base model. Their weights live in the same LoRA subspace.

We merge them by simple parameter averaging:

$$W_{\text{merge}} = \alpha W_{\text{AUX}} + (1 - \alpha) W_{\text{ALL}} \quad (4)$$

for each of the 504 LoRA tensors, with $\alpha = 0.5$ in our experiments. No fine-tuning. The merged adapter is loaded the same way as either parent.

This is task arithmetic Ilharco et al. [2023], Yadav et al. [2023] applied to game-theoretic specialists. We find empirically (§3.5) that W_{merge} matches the data-side intervention ($4\times$ upsampling auction+negotiation in the training mix) on aggregate metrics within sampling noise, suggesting the two specialists learned linearly composable subspaces rather than conflicting representations.

2.6 Pipeline summary

Algorithm 1 summarises one episode of $A+\beta$ data construction.

Algorithm 1 $A+\beta$ data construction (per episode)

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1: Input: game  $G$ , opponent policy  $\pi_{\text{opp}}$ , frontier paraphraser  $\text{LLM}_{\text{front}}$ , horizon-aware flag  $h$ 
2: Roll out blind trajectory  $\tau_{\text{blind}} = (x, \rho^{\text{blind}}, \mathbf{a}^{\text{blind}})$  with  $\text{LLM}_{\text{front}}$  as player 0
3: for round  $t = 0, \dots, T - 1$  do
4:    $a^* \leftarrow \text{BR}_0(a_t^1)$  ▷ Eq. 1
5:   if  $a^* \neq a_t^{0,\text{blind}}$  then
6:      $R_{\text{cf}} \leftarrow$  horizon-aware Eq. 3 if  $h$  else Eq. 2
7:     if  $R_{\text{cf}} > R(\tau_{\text{blind}})$  then
8:        $\rho_t^{\text{oracle}} \leftarrow \text{LLM}_{\text{front}}(\text{no-leak prompt}, a^*)$ 
9:       Emit pair  $(x_t, \rho_t^{\text{oracle}} \oplus a^* \succ \rho_t^{\text{blind}} \oplus a_t^{0,\text{blind}})$ 
10:    end if
11:  end if
12: end for
```

We train Qwen 2.5–3B with LoRA ($r = 16$, $\alpha = 32$, dropout 0.05) using the standard sigmoid DPO loss with $\beta = 0.1$ and environment-token loss masking (the small model is not penalised for matching env-emitted observation strings). Eval-loss-based best-checkpoint selection follows Rafailov et al. [2023]. Full hyperparameters in Appendix 5.

3 Experiments

3.1 Setup

Environments. 12-environment suite. **In-domain (8):** three iterated prisoner’s dilemma payoff variants (pd-classic $T = 5/R = 3/P = 1/S = 0$, pd-tight $T = 4/R = 3/P = 2/S = 1$, pd-high-temptation $T = 8/R = 3/P = 1/S = 0$, all $T_{\text{horizon}} = 10$), repeated stag-hunt, multi-issue bargaining (negotiation), Battle of Sexes, Matching Pennies, tic-tac-toe. **Held-out (4):** first-price sealed-bid auction (Uniform values, $v_{\text{max}} = 100$), divide-the-dollar, K -level p -beauty, one-shot prisoner’s dilemma stage (ipd-stage). For each (model, env) pair we run $n = 12$ episodes against a 9-opponent pool drawn per-environment.

Models evaluated. Six trained variants of Qwen 2.5–3B: base, hypothesis-B FILTER-ON, hypothesis-B FILTER-OFF, $A+\beta$ -CORE, $A+\beta$ -AUX, $A+\beta$ -ALL, $A+\beta$ -RW (data-side reweight), $A+\beta$ -MERGE (parameter-side LoRA average). All trained variants are Qwen 2.5–3B. Frontier reference rows: Bedrock Haiku 4.5, Gemma 3–27B, Llama 3.3–70B, Llama 3.1–8B. Qwen 2.5–3B base provides the untrained-backbone reference.

Table 1: Headline results. Qwen 2.5–3B trained variants vs. frontier references on 12-env eval (8 ID + 4 HO, $n = 12$ episodes per env). CR = mean cumulative reward. Reasoning-action coupling is reported separately in §3.7 (Table 5). Best 3B-variant cell per metric in **bold**.

Model	CR(ID)	CR(HO)
Qwen 2.5–3B base	14.04	−1.45
Qwen 2.5–3B + DPO (filter-on)	13.77	−0.95
Qwen 2.5–3B + DPO (filter-off)	13.51	−0.23
Qwen 2.5–3B + A+ β -CORE	13.79	+0.44
Qwen 2.5–3B + A+ β -AUX	13.86	+0.68
Qwen 2.5–3B + A+ β -ALL	14.34	−0.27
Qwen 2.5–3B + A+ β -RW	13.93	+1.91
Qwen 2.5–3B + A+ β -MERGE	13.92	+1.75
Bedrock Haiku 4.5	14.60	+2.05
Gemma 3–27B	13.36	+0.28
Llama 3.3–70B	12.29	+1.52
Llama 3.1–8B (Bedrock)	13.60	−0.06

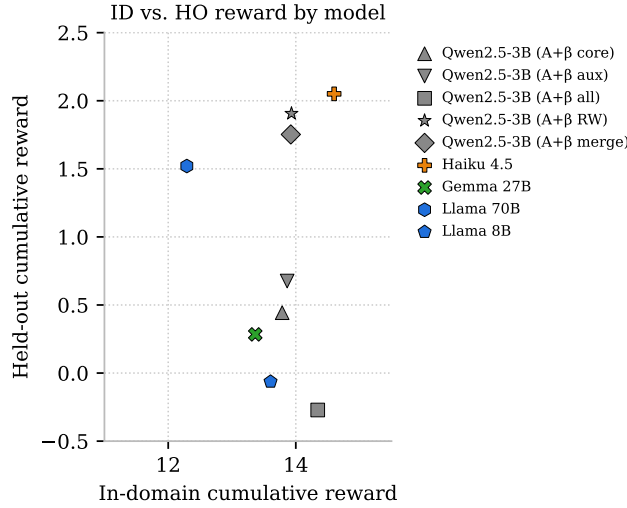


Figure 2: In-domain vs. held-out cumulative reward by model. Qwen 2.5–3B A+ β variants are shown in grey (distinguished by marker); frontier reference models are coloured (Haiku orange, Gemma green, Llama blue). A+ β -RW (star) and A+ β -MERGE (diamond) sit in the upper region near Haiku 4.5, recovering held-out reward without sacrificing in-domain reward.

197 **Metrics.** Cumulative reward (CR) per trajectory averaged across opponents, separately for in-
198 domain (ID) and held-out (HO) games. Reasoning-action coupling rate computed by parsing the
199 [EV] slot from chain-of-thought and comparing argmax-stated-EV to chosen action; reported as fc
200 (final-round) and ac (all-round). Parser is the prose-form-aware regex described in Appendix 5.

201 **Training.** Qwen 2.5–3B-Instruct with LoRA $r = 16$, $\alpha = 32$, dropout 0.05, 10 epochs, sigmoid
202 DPO loss, $\beta = 0.1$, env-token masking, learning rate 5×10^{-5} cosine. Best checkpoint by held-out
203 DPO eval-loss. Single L40S GPU, 13 GB peak. Training time scales linearly: 503 pairs (CORE) takes
204 60 min, 1338 pairs (ALL) takes 219 min.

205 3.2 Headline results

206 Table 1 summarises eight Qwen 2.5–3B variants and the four frontier reference models on aggregate
207 ID and HO cumulative reward; Figure 2 plots the same models in (ID, HO) reward space.

Table 2: Per-game cumulative reward. Trained 3B variants and Haiku 4.5 reference. Best 3B cell per row **bold**.

Game	base	core	aux	all	rw	merge	Haiku
<i>In-domain (ID)</i>							
pd-classic	21.96	22.08	23.08	23.88	21.67	23.50	25.68
pd-tight	25.92	27.08	26.21	26.21	25.42	25.92	27.25
pd-high-temptation	29.33	28.50	29.17	30.42	28.42	29.17	27.92
stag-hunt	28.62	28.71	28.38	30.54	30.42	27.88	27.79
negotiation	5.75	3.17	3.46	3.08	4.67	4.04	7.54
bos	0.75	0.75	0.62	0.62	0.88	0.88	0.62
matching-pennies	0.00	0.00	0.00	0.00	0.00	0.00	0.00
tic-tac-toe	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Held-out (HO)</i>							
auction	-8.67	-0.71	-0.10	-3.96	4.97	4.18	5.82
divide-dollar	0.27	0.11	0.22	0.34	0.32	0.21	0.25
p-beauty	0.50	0.50	0.50	0.50	0.50	0.50	0.50
ipd-stage	2.08	1.88	2.08	2.04	1.83	2.12	1.62

A+ β -RW and A+ β -MERGE close 86–93% of the held-out CR gap to Haiku 4.5. HO cumulative reward goes from base -1.45 to RW $+1.91$ and MERGE $+1.75$, against Haiku $+2.05$. No prior B-style training variant reaches positive HO CR. The gain is concentrated on auction (HO) and stochastic-opponent rounds (§3.4). Reasoning-action coupling, a complementary axis, is analysed separately in §3.7.

3.3 Per-game breakdown

Table 2 shows CR per game across all variants. RW and MERGE match or beat Haiku on three ID games (pd-high-temptation, stag-hunt, ipd-stage) but underperform on pd-classic (-2 CR) and negotiation (-3 CR). The auction gain (base $-8.67 \rightarrow$ RW $+4.97$, vs Haiku $+5.82$) is the largest single per-game improvement.

3.4 Per-opponent strategic-axis breakdown

Aggregating over all 12 envs, we group opponents by strategic axis (stationary vs. stochastic vs. strategic, see Table 3).

Stationary opponents (always-cooperate, always-defect): base already plays well (16.60 ID CR vs Haiku 18.79); training adds nothing substantial (RW 16.40).

Stochastic opponents (random, probabilistic): all 3B variants and Haiku cluster around 6–7 ID CR (base 7.22, RW 6.37, Haiku 6.83); training does not change the stochastic-opponent picture in-domain. (The held-out stochastic gain, where base loses heavily, is the relevant one — see Table 3.)

Strategic opponents (TFT, grim-trigger, pavlov): **this is where the lift concentrates.** Base 13.26 ID CR vs Haiku 21.13 (gap 7.87). RW reaches 14.62, MERGE 15.54. Per-opponent on TFT specifically, base is 1.50, RW is 14.06 ($+12.5$) — a near order-of-magnitude lift on the single most representative reciprocity opponent (Figure 3).

3.5 The AUX-vs-ALL trade-off and two interventions

Observation. Two earlier A+ β variants reveal a complementary failure pattern. AUX (613 pairs, 18% auction+negotiation) wins HO CR ($+0.68$) but loses ID strategic play (TFT-opp 12.88 vs ALL’s 15.31). ALL (1338 pairs, 56% horizon-aware-PD, 8% auction+negotiation) wins ID strategic play but the auction generalisation collapses (auction CR $-0.10 \rightarrow -3.96$). Diluting auction+negotiation share from 18% to 8% caused the auction forgetting.

Two interventions. (a) *Reweighted training* — upsample the 137 auction+negotiation pairs $4\times$ into a 1749-pair mix, retrain. (b) *LoRA averaging* — merge the AUX and ALL LoRA adapters by simple weight averaging (Eq. 4, $\alpha = 0.5$), no retraining.

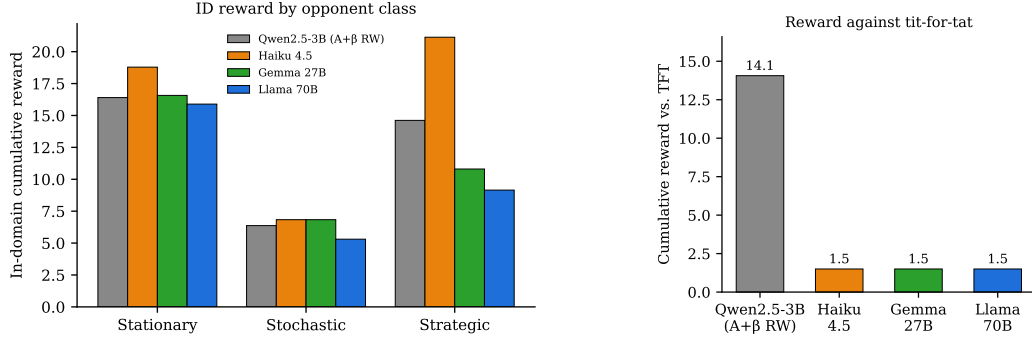


Figure 3: Left: in-domain reward by opponent class for the trained Qwen 2.5–3B A+ β -RW model against three frontier references (Haiku 4.5, Gemma 3–27B, Llama 3.3–70B). On stationary and stochastic opponents all four models are bunched; on *strategic* opponents A+ β -RW (14.62) clears the larger frontier models (Gemma 10.80, Llama-70B 9.15) and trails only Haiku (21.13). Right: reward against tit-for-tat specifically — A+ β -RW reaches 14.06, while all three frontier references score only 1.50.

Table 3: Per-opponent-axis CR (all 12 envs combined, $n = 12$ episodes per env). Opponents grouped by behavioural axis: stationary (always-C, always-D), stochastic (random, probabilistic), strategic (TFT, grim, pavlov, tit-greedy). **Strategic-opponent CR is where the lift concentrates:** base 13.26, A+ β -RW 14.62, Haiku 21.13.

Set	Axis	base	core	aux	all	rw	merge	Haiku
ID	Stationary	16.60	16.19	15.84	16.35	16.40	15.92	18.79
ID	Stochastic	7.22	5.53	6.66	6.21	6.37	6.34	6.83
ID	Strategic	13.26	15.12	15.25	16.35	14.62	15.54	21.13
HO	Stationary	1.41	1.29	1.48	1.48	1.37	1.49	1.10
HO	Stochastic	−3.33	−0.07	0.20	−1.32	2.23	1.91	2.67

239 **Result (Table 1).** Both interventions match each other within sampling noise. Per-tensor LoRA
240 averaging recovers combined capabilities that data-side reweighting also recovers, suggesting the two
241 specialists learned linearly composable LoRA subspaces.

242 3.6 Frontier reference points

243 The frontier reference rows are useful as a benchmark in their own right. Highlights: Llama 3.3–
244 70B has the worst aggregate strategic-opp CR (12.29) despite being the largest model. Haiku 4.5
245 dominates HO (2.05) but is mediocre on TFT specifically (1.50). The coupling disconnect for these
246 models is discussed in §3.7.

247 3.7 Reasoning-action coupling

248 Cumulative reward measures *what* a model plays; reasoning-action coupling measures whether its
249 stated reasoning predicts that play. We report final-round coupling *fc* (fraction of parsable rounds
250 whose chosen action matches the argmax of the stated [EV] slot) and all-round coupling *ac*. Table 5
251 gives both for the trained Qwen 2.5–3B A+ β variants and the frontier references; Figure 4 plots *fc* in
252 (ID, HO) space, mirroring the reward view of Figure 2.

253 **RW and MERGE recover the coupling the narrower mixes lost.** RW and MERGE reach ID *fc*
254 0.68 and 0.69, the best of any A+ β variant, while CORE through ALL dropped to 0.56–0.61. The
255 reweighting and merging interventions restore parsable reasoning that the narrower AUX and ALL
256 mixes had narrowed away, and they also lead on HO *fc* (0.46, 0.50). CORE is the cautionary extreme:
257 HO *fc* 0.00, reasoning fully decoupled from action on held-out games.

Table 4: Frontier reference rows on the same 12-env eval. Useful as a benchmark in itself. CR = mean cumulative reward; TFT-CR = cumulative reward against the tit-for-tat opponent. Coupling for these models is reported in Table 5.

Model	CR(ID)	CR(HO)	TFT-CR
Bedrock Haiku 4.5	14.60	+2.05	1.50
Gemma 3–27B	13.36	+0.28	0.75
Llama 3.3–70B	12.29	+1.52	1.50
Llama 3.1–8B (Bedrock)	13.60	−0.06	0.75
Qwen 2.5–3B base	14.04	−1.45	1.50

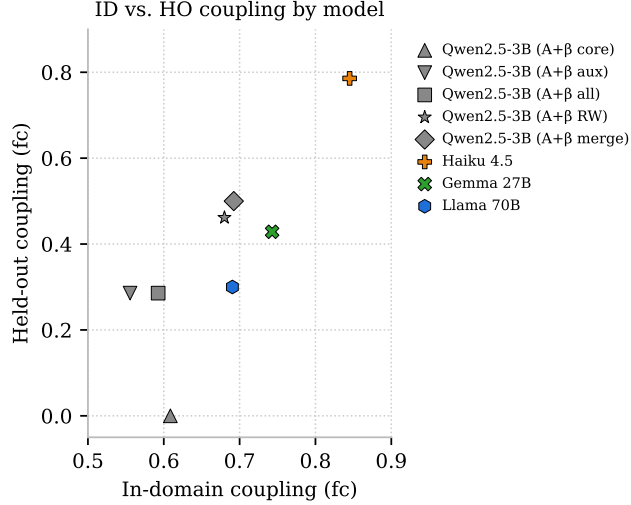


Figure 4: In-domain vs. held-out final-round coupling fc by model, in the same layout and colour scheme as Figure 2 (Qwen 2.5–3B $A+\beta$ variants grey; frontier references coloured). $A+\beta$ -RW (star) and $A+\beta$ -MERGE (diamond) sit highest among the 3B variants on both axes; Haiku 4.5 leads overall, while $A+\beta$ -CORE (triangle) collapses to zero held-out coupling.

258 **Frontier coupling is not aligned with frontier reward.** Gemma 3–27B has the highest all-round
259 ID coupling (0.80) but among the worst HO reward (CR +0.28, Table 1); Haiku 4.5 leads coupling and
260 reward together, but the two orderings otherwise diverge. Coupling and reward are complementary,
261 not redundant, axes.

262 3.8 Coupling-filter retrospective (Hypothesis B)

263 We initially hypothesised that DPO with a coupling filter would teach small models to bind reasoning
264 to action. The data falsified this: filter-on Qwen 2.5–3B fc(ID) went $0.77 \rightarrow 0.66$ (down), filter-off
265 went $0.77 \rightarrow 0.55$. CR(ID) flat-to-down. The retrospective remains a useful ablation: it shows that the
266 coupling-filter heuristic applied to frontier-action labels is the wrong intervention — the intervention
267 that works ($A+\beta$) replaces the action label, not just the filter.

268 4 Related Work

269 **LLMs in non-cooperative games.** Recent benchmarks have established that LLMs underper-
270 form game-theoretic equilibrium concepts in non-cooperative settings beh [2025], gtb [2024], gam
271 [2024b,a]. non [2024] surveys the literature on LLM strategic reasoning failures.

272 **Direct preference optimization.** $A+\beta$ builds directly on DPO Rafailov et al. [2023], with
273 environment-token loss masking implemented as a custom MaskedDPOTrainer subclass on top
274 of TRL von Werra et al. [2020]. We considered SimPO Meng et al. [2024] and IPO Azar et al. [2024]

Table 5: Reasoning-action coupling on the 12-env eval. fc = final-round coupling (fraction of parsable rounds whose chosen action matches the argmax of the stated [EV]); ac = all-round coupling. Reported for the trained Qwen 2.5-3B A+ β variants and the frontier references. Best 3B-variant cell per metric in **bold**.

Model	fc(ID)	ac(ID)	fc(HO)
Qwen 2.5-3B + A+ β -CORE	0.61	0.74	0.00
Qwen 2.5-3B + A+ β -AUX	0.56	0.72	0.29
Qwen 2.5-3B + A+ β -ALL	0.59	0.65	0.29
Qwen 2.5-3B + A+ β -RW	0.68	0.70	0.46
Qwen 2.5-3B + A+ β -MERGE	0.69	0.74	0.50
Bedrock Haiku 4.5	0.85	0.57	0.79
Gemma 3-27B	0.74	0.80	0.43
Llama 3.3-70B	0.69	0.52	0.30

as alternative loss formulations but did not find them necessary; standard sigmoid loss with $\beta = 0.1$ trained stably. liu [2024] surveys preference-optimization variants relevant to our recipe.

Process supervision and credit assignment. Process reward models Lightman et al. [2024] assign per-step credit to chain-of-thought tokens. Our approach is closer to step-level DPO with the wrinkle that the “correct step” is the analytical solver’s best response rather than a learned reward model’s score. GRPO Shao et al. [2024] solves the same credit-assignment problem in sequential settings via K on-policy rollouts and group-relative normalisation. Our horizon-aware counterfactual (Eq. 3) replaces this K -sample estimate with a single deterministic dynamic-programming pass when the opponent’s policy is known and reconstructible. The tradeoff is exactness vs. generality.

Model merging and task arithmetic. Our specialist-adaptor LoRA averaging (§2.5) is task arithmetic Ilharco et al. [2023] applied to game-theoretic specialists. Yadav et al. [2023] (TIES-merging) and Matena and Raffel [2022] provide more sophisticated merging schemes; we find that simple weight averaging suffices for our use case. To our knowledge, this is the first demonstration of LoRA-merge equivalence specifically for strategic-game DPO specialists.

Multi-agent RL with LLMs. owl [2025], mas [2024b,a] demonstrate cooperative multi-agent LLM systems. Adversarial multi-agent settings remain underexplored adv [2024], Anthropic [2024]. Our work focuses on the 2-player non-cooperative case as a tractable empirical entry point.

5 Discussion and Limitations

What A+ β closes. On a 12-environment suite, the A+ β -trained Qwen 2.5-3B closes 93% of the held-out cumulative-reward gap to Haiku 4.5 (+1.91 vs +2.05, base -1.45), lifts TFT-opponent reward from 1.50 to 14.06, and matches/beats Haiku on three of eight in-domain games. The held-out gain is concentrated on the auction environment (base -8.67 $\rightarrow$ RW +4.97).

What A+ β does not close. Aggregate ID coupling 0.68 vs Haiku 0.85 remains. On games where Haiku exploits stationary opponents particularly well (pd-classic against always-cooperate), the trained model loses 1-2 reward points. Negotiation in-domain training did not generalise as cleanly as auction. Solver coverage limits us to game families with tractable analytical best responses; extending A+ β to e.g. tic-tac-toe (sequential perfect-information, minimax) or open-ended dialogue games is open work.

Generality of LoRA averaging. The specialist-adaptor merge result (§2.5) suggests that adapters trained with the same DPO recipe but different data slices can be combined without retraining, but we have demonstrated this for two adapters at $\alpha = 0.5$ on a single base model (Qwen 2.5-3B). Extending to N specialists, asymmetric weights, or cross-base-model merges is future work.

Conclusion. Frontier LLMs and small base LLMs both play poorly against strategic opponents in 2-player non-cooperative games. A targeted DPO recipe A+ β that replaces the action label

309 with an analytical solver’s best response, paraphrases reasoning under a no-leakage prompt, and
310 filters by counterfactual horizon rollout closes most of the held-out reward gap to the frontier and
311 lifts strategic-opponent reward by an order of magnitude. The gain generalises out-of-domain to a
312 structurally different game (auction) that we did not train on with the same intensity. Two specialist
313 adapters trained on disjoint data slices merge by simple parameter averaging to recover combined
314 capabilities. We hope the 12-env benchmark and method recipe are useful entry points for further
315 investigation of single-agent RL recipes that produce multi-agent-aware LLM agents.

316 Appendix

317 A. Hypothesis B retrospective: why coupling-filter DPO didn't work

318 Before $A+\beta$, we ran a full DPO pipeline (which we call Hypothesis B internally) that pinned chosen-
319 side actions to the frontier model's recorded play and filtered pairs by reasoning-action coupling. The
320 recipe was:

- 321 1. Generate trajectories with frontier (Haiku 4.5) as player 0, heuristic opponents.
- 322 2. For each trajectory, parse the [EV] slot in chain-of-thought. Tag a trajectory "coupled" if the
323 played action equals the argmax of the parsed EVs.
- 324 3. Build DPO pairs by S1 (outcome / cumulative reward), S2 (NE-aligned action), S3 (cross-
325 strategy consensus), S4 (L2 vs L0 contrastive). Filter chosen-side to coupled-only trajecto-
326 ries.

327 The headline coupling claim — that filtering teaches small models to bind reasoning to action — did
328 not survive the data. Filter-on Qwen 2.5–3B post-DPO final-round ID coupling went from 0.77 (base)
329 to 0.66 (down). Filter-off went $0.77 \rightarrow 0.55$ (further down). Two cumulative reward observations
330 were also off-prediction: filter-off (-0.23 HO CR) beat filter-on (-0.95), and both lost ID CR vs
331 base. We had predicted the opposite ordering.

332 **Diagnosis: training-distribution narrowness.** Five of the eight ID training games (PD $\times$ 3,
333 stag-hunt, BoS) sit on a single "cooperate-or-defect-vs-reciprocity" axis. Three of those five are
334 the same matrix-game stage with different payoff parameters. The opponent pool is similarly
335 concentrated: TFT, AlwaysCoop, AlwaysDefect, GrimTrigger, Pavlov, TitForTwoTats, GenerousTFT,
336 EpsilonGreedy — seven of nine are reciprocity-style. Under such a narrow training mix, DPO
337 overwrites whatever decision habits the base RLHF gave Qwen 2.5-3B with stricter reciprocity
338 patterns. The chosen-side coupling filter narrows the support further (388 pairs vs 407 unfiltered),
339 giving filter-on *less* variety to learn from — the opposite of what the filter was supposed to do.

340 This diagnosis bridged us to $A+\beta$. Filter-off's modest HO CR gain ($+1.22$ CR vs base) suggested
341 wider chosen-side distributions generalise better. $A+\beta$ replaces the filter with a different intervention:
342 rather than narrowing chosen-side trajectories to "frontier action that happens to match its own
343 argmax-EV," it *replaces the action label entirely* with the analytical solver's best response. The data
344 is then unrelated to the filter mechanism; the question becomes whether better action labels transfer
345 better than filtered frontier-pinned action labels. The answer (§3) is yes.

346 **Why we report this in the paper.** Two reasons: (i) the falsified coupling thesis was load-bearing in
347 our initial framing, and reporting only the working method without the prior failure hides where the
348 actual contribution sits; (ii) the diagnostic claim about narrow training distributions causing forgetting
349 is itself a reusable observation — it returned in the AUX-vs-ALL trade-off (§3.5), where it was again
350 resolved by widening the chosen-side distribution rather than narrowing it.

351 B. Frontier-coupling measurements

352 Frontier final-round coupling rates by environment, computed by the prose-form-aware parser
353 (Appendix 5). n/a indicates the env had no parsable EV slots in any final round (typical for tic-tac-toe
354 where the prompt does not elicit slot form, and for continuous-action envs).

355 Two patterns visible. (i) Coupling drops sharply on held-out games for all but Haiku and Gemma. (ii)
356 Llama 3.3-70B has the largest *aggregate* parameter count but lower PD-game coupling than Gemma
357 3–27B and Llama 3.1–8B — scale alone does not explain coupling.

358 C. Leakage audit

359 The $A+\beta$ frontier paraphrase prompt (§2.3) is engineered to write reasoning that does not reference
360 the current-round opponent action as observed. We audit two prompt versions: v1 (initial design, told
361 frontier the opponent's move directly), v2 (no-leakage rewrite, posterior-EV framing).

Table 6: Per-(model, env) final-round coupling rate, parsable rounds only. $n = 12$ episodes per cell. Game abbreviations: pd-c = pd-classic, pd-t = pd-tight, pd-h = pd-high-temptation, mp = matching-pennies, ttt = tic-tac-toe, auct = auction, dd = divide-dollar, p-b = p -beauty, ipd = ipd-stage.

Model	pd-c	pd-t	pd-h	stag	nego	bos	mp	ttt	auct	dd	p-b	ipd
Haiku 4.5	0.87	0.90	0.91	0.89	0.00	1.00	1.00	n/a	0.00	n/a	0.00	0.92
Llama 3.1–8B (B)	0.90	0.43	1.00	0.43	0.00	1.00	0.71	0.00	n/a	0.00	n/a	0.67
Gemma 3–27B	0.38	0.75	0.83	0.91	0.00	1.00	0.91	0.00	n/a	0.00	n/a	1.00
Llama 3.3–70B	0.60	0.33	0.80	0.50	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.75
Qwen 2.5–3B base	1.00	0.80	0.83	0.75	n/a	0.78	0.75	0.00	n/a	0.00	0.00	1.00
Qwen 2.5–3B + A+ β -on	0.83	0.33	0.50	1.00	0.00	1.00	0.60	n/a	0.00	0.00	0.00	0.67
Qwen 2.5–3B + A+ β -off	0.25	0.33	0.33	0.00	0.00	0.88	0.88	0.00	n/a	n/a	0.00	0.83

v1 leakage rate. On a sample of 4 round-1 oracle traces across 7 envs, all 4 round-1 traces wrote phrases like “The opponent played C in round 1” or “opponent played Stag this round”. v1 was unsafe to train on: the small model would learn to hallucinate observed-this-round actions at deployment.

v2 leakage rate. Across all 688 oracle rounds in the final batches (112 round-1 + 576 later rounds), the leakage regex `r"opponent (played|chose|cooperat|defect)\w*\s+(this round|in round \d+)"` matches 7 cases. Manual inspection confirms all 7 are false positives discussing prior rounds (e.g. “Round 1 history: I played C, opponent played C”). The round-1-specific regex `r"opp(onent)? play(ed|s)|opp(onent)? cooperat|opp(onent)? defect"` matches 6 round-1 traces; 5 are false positives discussing hypothetical opponent types (“opponent could be a cooperator”), 1 is a model echoing the system prompt’s own prohibition.

Empirical conclusion. v2 leakage rate is effectively 0. The +1.91 HO CR gain from A+ β -RW is not explained by hindsight hallucination at training time.

D. Coupling parser

We parse chain-of-thought for an [EV] slot and extract per-action expected values via three regex patterns. All patterns require an action label, an arrow/colon/equals separator, an arithmetic expression with at least one operator, and a final equals-result number.

- **Pattern A** (slot form): bare action label + colon or arrow. Examples: “C: 0.5*3 + 0.5*0 = 1.5”, “Stag → 0.5*4 = 2.0”, “**Action C:** 0.5*3 = 1.5”.
- **Pattern B** (EV-wrapper form): “EV(C) = 0.30(3.0) + 0.40(0) = 1.2”. Common in Anthropic outputs.
- **Pattern C** (prose form, post-DPO): “Expected payoff for ‘C’ is 0.6*3 + 0.4*0 = 1.8”, “Total EV for ‘D’ = 1.5 + 0.7 = 2.2”. Trained Qwen models tend to produce this form.

After regex match, an action label is dropped if it appears in a list of English-prose words mistakenly captured (round, rounds, value, playing, etc.).

A round is *coupled* iff the action with the maximum stated EV equals the chosen action. A round is *parsable* if at least one EV expression matches; rounds without parsable EVs are excluded from the denominator. Final-round coupling restricts to the last round of each episode.

Released as `rl_noncoop/scoring/coupling.py` alongside 17 unit tests (matrix-game stages, prose-form Qwen outputs, edge cases). Adding Pattern C lifted parsable-round counts on Qwen 2.5–3B PD outputs by $\sim 3\times$ on average, exposing real coupling rates that the slot-only parser had missed.

E. Training hyperparameters

All trained A+ β variants use Qwen 2.5–3B-Instruct, LoRA with $r = 16$, $\alpha = 32$, dropout 0.05, target modules auto, max sequence length 12288. Training: 10 epochs, per-device batch 1, gradient accumulation 8 (effective batch 8), learning rate 5×10^{-5} cosine schedule with 5% warmup, gradient

checkpointing on, no flash attention, sigmoid DPO loss with $\beta = 0.1$, environment-token loss masking on, save every 10 steps, `load_best_model_at_end=true`. Single L40S GPU, 13–14 GB peak memory. The Hypothesis B variants share the same recipe.

Variant	Pairs	Best ckpt step	Best eval loss
B filter-on	388	60	0.3734
B filter-off	407	110	0.2721
A+ β -CORE	503	50	0.2973
A+ β -AUX	613	80	0.3128
A+ β -ALL	1338	40	0.2278
A+ β -RW	1749	70	0.2454
A+ β -MERGE	—	—	—

Table 7: Training hyperparameters consistent across all DPO variants except for pair count and best checkpoint. A+ β -MERGE is not a trained variant; it is the LoRA-average of AUX and ALL adapters.

F. Two example A+ β pairs

We include two pedagogical examples from `runs/oracle_batch1_*` to illustrate the kind of training signal A+ β produces.

Example 1 — pd-classic, episode 0, round 10 (final round). Iterated PD with $T = 5/R = 3/P = 1/S = 0$, opponent = TFT. Both sides cooperated for 9 rounds. Round 10 is the final round; no shadow of the future. Blind player 0 played C (cumulative reward 30.0) despite computing $EV(D) = 4.95 > EV(C) = 3.0$ in its own [EV] slot.

Blind reasoning [Decision] excerpt:

[DECISION] Play C. While defection yields 5 vs 3 this round, the opponent’s 9-round perfect cooperation record and the statement that “the game continues” suggest either: (1) there are more rounds coming, or (2) reputation matters. Defecting would break a mutually beneficial pattern.

A+ β oracle reasoning [EV] and [Decision] excerpts (counterfactual reward 32.0, +2):

[EV] This is the final round ($T=10$), so there is no future to consider. C: 3.000 (mutual cooperation payoff). D: 5.000 (defection payoff when opponent cooperates).
[DECISION] Play D. In the final round of a finite repeated game, the standard solution is to defect regardless of history; the opponent cannot retaliate.

The blind trace’s reasoning was *coupled-internal* (it computed EVs and chose argmax) until the [Decision] step — where it overrode the EV with a horizon argument that does not apply to the last round. Hypothesis B’s coupling filter would not have caught this: the EV computation matched the chosen action up to the very last sentence. A+ β ’s solver says final-round-stage-best-response is D, the counterfactual confirms +2 CR over what blind achieved, and the pair is emitted with the oracle excerpt as chosen and the blind excerpt as rejected.

Example 2 — stag-hunt, episode 2, anti-coordination. 10-round stag-hunt with payoffs (Stag,Stag)= 4, (S,H)= 0, (H,S)= 3, (H,H)= 1, opponent = TFT. Blind player 0 was *anti-coordinating* with TFT (Table 8). Blind cumulative reward was 19.0; A+ β ’s oracle would have achieved 24.0 across the 5 helps-flagged rounds.

Blind’s [Prior] for round 1 placed 0.4 probability on an exploitative opponent type, leading to a Hare-as-risk-dominant decision. The actual opponent (TFT) was zero-probability exploitative. This is the failure mode Hypothesis B’s coupling filter literally cannot reach: blind’s EV-action coupling for that round was 1.0 (it chose argmax of its own stated EVs); the mistake was one level up, in the prior. A+ β factors out prior-formation by giving the small model a solver-pinned answer: against TFT-played-Stag, best response is Stag. The small model learns from supervision; it does not have to derive the prior from scratch.

round	TFT	blind	oracle
1	Stag	Hare	Stag (HELPS)
2	Hare	Stag	Hare (HELPS)
3	Stag	Hare	Stag (HELPS)
6	Hare	Stag	Hare (HELPS)
7	Stag	Hare	Stag (HELPS)

Table 8: Stag-hunt anti-coordination. Blind p0 played the opposite of TFT’s previous action each round. Other rounds (4, 5, 8, 9, 10) agreed.

435 G. Per-(model, env) coupling, exploitability, OPA

436 The full per-cell breakdown is large (96 rows: 8 reference models + DPO variants $\times$ 12 envs).
437 We reproduce the Haiku 4.5 row here to anchor the reader; the complete table is available in the
438 companion repository.

Haiku 4.5 / env	ac	fc	exploitability
pd-classic	0.44	0.91	0.73
pd-tight	0.47	0.90	0.72
pd-high-temptation	0.54	0.91	0.58
stag-hunt	0.85	0.89	0.27
negotiation	0.00	0.00	n/a
bos	1.00	1.00	0.00
matching-pennies	1.00	1.00	1.00
tic-tac-toe	0.00	n/a	n/a
auction	0.00	0.00	n/a
divide-dollar	n/a	n/a	n/a
p-beauty	0.00	0.00	n/a
ipd-stage	0.92	0.92	0.00

Table 9: Haiku 4.5 per-env companion metrics. ac = all-round coupling, fc = final-round coupling. Exploitability via maxmin LP for matrix games; n/a for continuous-action and one-shot games where the metric is undefined or trivially zero.

439 H. Opponent pools by environment family

440 **Repeated 2×2 matrix games (PD $\times$ 3, stag-hunt).** 9 deterministic heuristics with weighted
441 sampling: TFT (weight 2.0), GrimTrigger (2.0), TitForTwoTats (1.5), GenerousTFT (1.5), Pavlov
442 (1.5), AlwaysCooperate (1.0), AlwaysDefect (1.0), RandomPolicy (1.0), EpsilonGreedy (1.0). Per-env
443 opponent sub-selection in our YAMLS.

444 **First-price auction.** Four bidder heuristics with round-robin warm-up sampling:
445 truthful_bidder ($b=v$), shaded_bidder_low ($b=0.3v$), shaded_bidder_high ($b=0.7v$),
446 random_uniform_bidder ($b \sim \text{Unif}[0, v]$).

447 **Multi-issue bargaining.** Four bargainer heuristics with round-robin warm-up:
448 greedy_bargainer (claims capacity on every issue), equal_split_bargainer (claims
449 half capacity each), nash_bargainer (claims all of own highest-weight issue, half elsewhere),
450 random_bargainer (uniform random proposal in $[0, c]^3$).

451 **Round-robin warm-up.** For all single-shot game families, we replaced the original purely-
452 weighted-random opponent sampler with a round-robin warm-up that ensures every opponent receives
453 at least one episode in the first $|\text{pool}|$ draws. This was a fix for a sampler bug discovered in our
454 second batch where TFT drew zero episodes despite a non-zero weight; the round-robin guarantees
455 minimum coverage at small batch sizes.

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